

TWINKIT · DESIGN PRECIS

TwinKit design precis

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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REVISION HISTORY

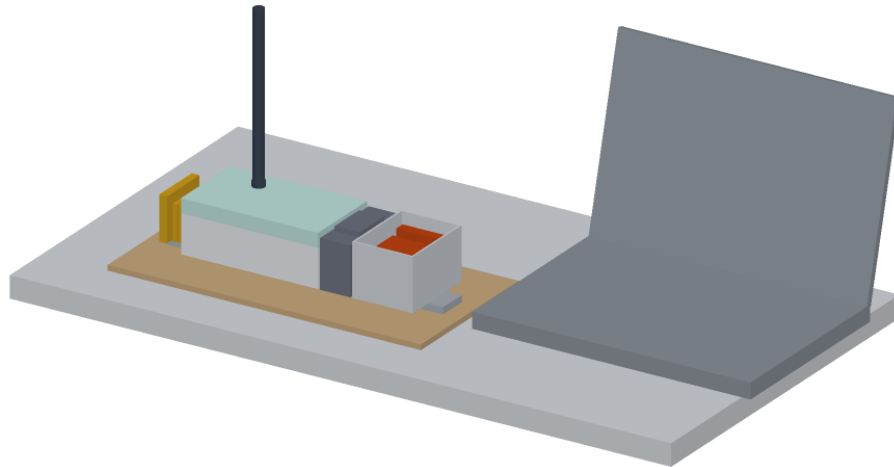
Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, safety, media)
0.1	2026-09-25	Amish Chadha	Initial scaffold

TwinKit design precis

TwinKit is a small DIN rail edge gateway plus open software that receives sensor readings, stores them, and shows each reading on the project's own build123d model next to the value the design calculation predicted. First-order numbers suggest one single-board computer with a LoRaWAN concentrator can serve 50 sensor nodes, store a year of data and ride through a 2 h outage on about 6 W. Parts cost about \$285, above the \$200 budget; options are listed below and in REVIEW.md.

TwinKit

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Grey: Desk, 14-inch laptop for scale

Figure 1. Gateway on a DIN rail and bench plate, with a desk and 14-inch laptop for scale.

■ How it works

1. **Sense.** Sensor nodes such as FieldNode (a sibling lab project) send small readings over LoRa radio. Other devices can post over MQTT or HTTP on the local network.
2. **Receive.** An 8-channel LoRaWAN concentrator on the gateway hears all nodes in range; an open-source network server decodes the packets and publishes them to a local MQTT broker.
3. **Store.** A time-series database on the gateway keeps every reading with its node, channel, unit and time.

4. **Map.** A mapping file for each project (the twin file) links every channel to a named part of the project's build123d model, using the same part names and BOM numbers the concept media already use, and to an expected value: a constant, a curve, or a function from the project's calculation note.
5. **Compare.** The TwinKit twin service computes the residual (measured minus expected) for each reading and raises a flag when it stays outside the stated tolerance for the stated time.
6. **Show.** A browser dashboard shows trends and flags, and a 3D view loads the project's `model.glb` and colors each mapped part by its live value or residual.

TwinKit: data flow from sensor to twin (values are estimates)

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Figure 2. Data flow from sensor to twin. Values are estimates.

An illustrative twin file for the proposed first example (FieldNode) might read:

```

project: fieldnode
model: media/model.glb
channels:
  - id: battery_v
    node: fieldnode-01
    part: 4          # BOM number of the LiFeP04 cell in FieldNode's model
    unit: V
    expected: calc.power_budget.soc_to_voltage  # from FieldNode's calculation note
    tolerance: 0.1
    hold: 30 min
  
```

The schema is a concept sketch; its fields are proposed, awaiting Amish.

■ Main components

Table 1. Main components (numbers match `bom/bom.csv` and Figure 3).

#	Component	Proposed choice	Notes
1, 2	Bench plate and DIN rail	6 mm plate, TS35 rail	Plate optional when an equipment cabinet exists
3, 4	Gateway enclosure	9-module DIN enclosure, vented, clear cover	Holds the computer and concentrator
5, 6	Single-board computer and cooler	4 GB quad-core Arm board (Raspberry Pi 5 class)	Runs all software. Proposed, awaiting Amish
7	LoRaWAN concentrator	SX1302 or SX1303 8-channel HAT	Region version (EU868 or US915). Proposed, awaiting Amish
8	Storage	64 GB high-endurance microSD	SSD upgrade is an option
9	Antenna	3 dBi whip on an SMA bulkhead	Mounted on the cover or outside a cabinet
10	DC-DC converter	9 to 30 V in, 5.1 V 5 A out	Accepts a 12 V adapter or an existing 12 V system
11, 12	Backup	DIN 12 V UPS module with a 12.8 V, 1.5 Ah LiFePO4 pack	Power-fail signal triggers a clean shutdown
13	Terminals and fuse	DIN terminal blocks, 2 A fuse	Fuse at the supply input
14	Open-source stack	LoRaWAN network server, MQTT broker, time-series database, dashboard	All open source, no cloud account
15	Twin service and schema	Python service and YAML twin file	TwinKit's own code, MIT
16	Example twin	FieldNode battery and solar charge	Proposed, awaiting Amish

TwinKit: exploded view

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- 1 Bench mounting plate
- 2 DIN rail, TS35
- 3 Gateway enclosure base
- 4 Gateway enclosure cover
- 5 Single-board computer, 4 GB
- 6 Active cooler
- 7 LoRaWAN concentrator HAT
- 8 microSD card, high endurance
- 9 LoRa antenna and SMA bulkhead
- 10 DC-DC converter, 12 V to 5 V
- 11 DIN UPS module
- 12 LiFePO4 backup pack, 12.8 V
- 13 Terminal blocks and fuse

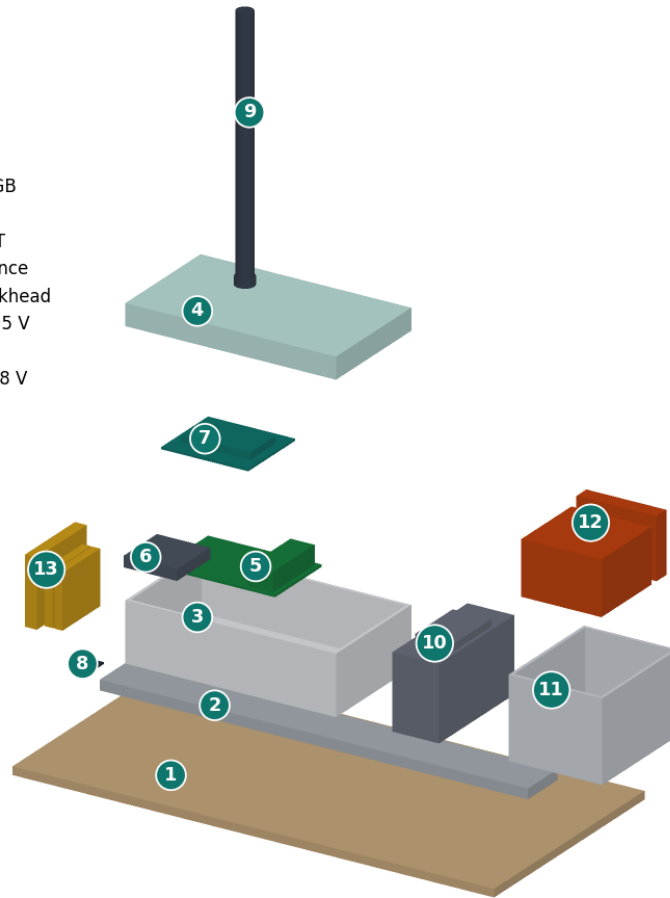


Figure 3. Exploded view with BOM numbers.

TwinKit: cutaway

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Figure 4. Cutaway through the gateway enclosure (left), converter (center) and UPS with backup pack (right).

■ First-order numbers

All values are estimates for concept review and will be checked at TRL 3.

Table 2. First-order numbers.

Quantity	Estimate	Basis	Requirement
Uplinks handled	600 per hour	50 nodes x 12 readings per hour	R1
Uplink loss from collisions	under 0.3 % at SF7; about 5 % if all nodes use SF12	Pure ALOHA, 75 uplinks per channel per hour, 60 ms or 1.3 s airtime	R1 met at SF7 to SF9
Stored values per year	about 53 million	50 nodes x 10 fields x 288 per day x 365	R3
Storage per year	about 1.6 GB	30 bytes per value before compression	R3 met (64 GB card)
Reading to dashboard	under 5 s	MQTT and database writes in well under 1 s each; dashboard refresh 2 s	R5 met (60 s)
Average power	about 6 W (0.5 A at 12 V)	Computer about 4 W, concentrator about 1 W, losses about 1 W	R8 met (8 W)
Backup time	about 2 h	19.2 Wh x 80 % usable, 90 % converter efficiency, 6 W load gives 2.3 h	R9 met (30 min)
Enclosure air temperature rise	about 20 K	6 W over about 0.059 m ² of enclosure surface at 5 W/(m ² ·K)	R12 not yet shown
Rail length used	about 300 mm (17 modules)	9 + 2 + 4 modules plus terminals	R10 met
Parts cost	about \$285	Indicative prices, see <code>bom/bom.csv</code>	R14 not met (\$200)

The thermal estimate is the main technical risk: at 40 °C ambient, enclosure air near 60 °C leaves little margin before the processor throttles. Vents, the active cooler and a light software load are expected to be enough, but this must be checked at TRL 3.

■ Key design choices

- **Compare with the calculation, not only with thresholds.** Most dashboards alarm on fixed limits. TwinKit compares each reading with what the design said it should be at that moment, so a battery that charges more slowly than the power budget predicted is caught before it goes flat.
- **Reuse the portfolio's own model outputs.** Every lab repo already exports a `model.glb` with named, BOM-numbered parts. The twin file points at those names, so no separate 3D model is built for the twin.
- **Edge first.** Everything runs on the gateway. An upstream server is optional, which keeps the kit usable where connectivity is poor and keeps data with its owner.
- **LoRaWAN rather than a single-channel receiver.** An 8-channel concentrator follows the open standard and accepts third-party sensors, at about \$55 more than a point-to-point receiver. Proposed, awaiting Amish.
- **Lightweight stack over a twin platform.** Frameworks such as Eclipse Ditto assume a server cluster; a single-board computer running a network server, MQTT, a time-series database and a small Python service fits in 4 GB. Proposed, awaiting Amish.
- **Monitoring only.** No control outputs, so a software fault cannot move a pump or switch a load.

■ Relation to other lab projects

- **FieldNode** is the primary data source and the proposed first example twin.
- **CityTwin** builds its public map and kiosk on TwinKit's gateway and software.
- **CellGuard** could protect a larger backup pack in a later version; at this size an off-the-shelf pack with a built-in BMS is proposed.
- **CalRig** calibration records could feed each channel's tolerance in the twin file.

■ Safety

Safety: The backup pack is a lithium (LiFePO₄) battery. Use a pack with a built-in BMS and fuse, charge it only through the UPS module within the maker's limits, keep it away from heat, and never leave a first build charging unattended.

Safety: The gateway runs from 12 V DC. Use a certified, double-insulated mains adapter; any wiring to an existing 12 V system or into a mains cabinet must be done or checked by a qualified electrician under local code.

Safety: A network gateway is an attack surface. Change every default credential, keep it off the public internet until it is hardened, apply updates, and treat sensor data as the owner's data.

Safety: TwinKit is a monitoring aid. It must not be relied on as a safety system, and it does not replace alarms, protection devices or inspections that a regulation or the equipment maker requires.

■ Open questions for TRL 3

- Thermal check of the enclosure at 40 °C ambient; do vents suffice or is a larger enclosure needed?
- Confirm airtime and loss figures for the chosen spreading factors and FieldNode's payload format.
- Choose the time-series database (TimescaleDB, InfluxDB or SQLite) against memory use and card wear.
Proposed: TimescaleDB, awaiting Amish.
- Define the twin file schema and how expected values are imported from a project's calculation note.
- microSD wear over a year of writes, or move to an SSD.

Concept media: [blueprint sheet](#), [interactive 3D model](#).